

One small-field blocking step, part 1b: the decay rate and the shape of the threshold, and why the weak side's deficit is ten orders of magnitude

Part 1 (Gaussian level) left one constant unresolved, the exponential decay rate κ of the fluctuation propagator, and part 2 was to produce the Kotecký–Preiss threshold g_{pert}^2 of the remainder gas. This note settles what can be settled about both without new mathematics, and the result reshapes the weak-side programme. **The literature carries no explicit rate.** Dimock's exposition of Balaban's construction (Rev. Math. Phys. 25 (2013) 1330010, §2.4; read from the arXiv text) proves the decay by a parametrix built from Neumann inverses on cubes of side M and a random-walk expansion that converges “for M sufficiently large”, with constants “depending on L but no other parameters”; the local estimate (80) there has a rate γ_0 stated only as $O(L^{-2})$. **Explicit routes give $\kappa \sim 10^{-3}$.** The generic Combes–Thomas argument gives 7×10^{-4} for the hard constraint (part 1); the soft-constraint operator $-\Delta + aQ^T Q$ is local, has gap $\frac{2}{9}$ for $a \geq 130$, and gives 1.7×10^{-3} ; the analytic-structure route through the folded symbol is set up in Section 2 and reduces to a zero-free strip for one explicit function, which is the defined computation for an order-one rate. **The threshold scales as κ^4 .** With the tree-graph organization of the remainder gas,

$$g_{\text{pert}} \simeq \frac{1}{e C_3 c_{\text{adj}} \|C_{\text{fl}}\|_{1,\kappa}/g^2} \simeq \frac{\kappa^4}{2 \times 10^4 C_3},$$

with C_3 the cubic-vertex constant, of order 10 to 30, so that even at the expected order-one rate $\kappa \simeq \frac{1}{2}$ the threshold is $g_{\text{pert}}^2 \sim 10^{-12}$, and at the proven rate it is 10^{-35} . Against the physical onset of the perturbative running at $g^2 \simeq 1$, **the weak side's explicit deficit is twelve orders of magnitude in g^2 with an ideal propagator and more than thirty with the proven one, where the strong side's is a factor 400 in β_W .** The sub-target H1a of the research-directions note is therefore closed as a constant-chasing problem: no sharpening of the present method reaches $g^2 \sim 1$, and H1 is a methods problem. The estimate of the threshold is a scaling estimate with explicit but crude constants, labelled as such; the decay-rate statements are proved. Constants explicit; nothing promoted.

1. What the literature does, and the soft constraint

Dimock's Green's function is $G_k = (-\Delta + \bar{\mu}_k + a_k Q_k^T Q_k)^{-1}$: the block averaging enters as a **soft** Gaussian constraint $aQ^T Q$ rather than the hard conditioning of part 1, and the operator is then local. The decay is proved by covering the lattice with cubes of side M , inverting with Neumann conditions on each, gluing with a partition of unity, and expanding the error in a random walk whose steps carry a factor $O(M^{-1})$; the walk converges for M large and the rate is half the local rate. No numerical value is attached to M , C or γ_0 anywhere in the construction, and Balaban's original (Commun. Math. Phys. 95 (1984) 17) is of the same kind.

The soft constraint can be made explicit here. With Q the averaging of part 1 (normalized by $\frac{1}{16}$), $QQ^T = (48 + 8(S + S^{-1}))/256$ along μ has spectrum in $[\frac{1}{8}, \frac{1}{4}]$, so $\|Qf\|^2 \geq \frac{1}{8}\|f_{\perp}\|^2$ for the component $f_{\perp}$ of f orthogonal to $K = \ker Q$. Writing $f = f_K + f_{\perp}$ and using $\|\partial f\|^2 \geq \frac{1}{2}\|\partial f_K\|^2 - \|\partial f_{\perp}\|^2$ with $\|\partial f_K\|^2 \geq \frac{4}{9}\|f_K\|^2$ (part 1, Proposition 1) and $\|\partial f_{\perp}\|^2 \leq 16\|f_{\perp}\|^2$,

$$\langle f, (-\Delta + aQ^T Q)f \rangle \geq \frac{2}{9}\|f_K\|^2 + \left(\frac{a}{8} - 16\right)\|f_{\perp}\|^2 \geq \frac{2}{9}\|f\|^2 \quad (a \geq 130).$$

The operator has range 2 along μ and 1 transversally, with $\|Q^T Q\| \leq \frac{1}{4}$. Conjugating by $e^{\kappa x \cdot e}$ changes $-\Delta$ by at most $2 \sinh \kappa$ and $aQ^T Q$ by at most $\frac{a}{4}(e^{2\kappa} - 1)$, so the conjugated operator stays invertible with norm ≤ 9 when $2 \sinh \kappa + \frac{130}{4}(e^{2\kappa} - 1) \leq \frac{1}{9}$, that is

$$\kappa \leq 1.7 \times 10^{-3}, \quad |(-\Delta + aQ^T Q)^{-1}(x, y)| \leq 9 e^{-\kappa|x-y|_{\infty}}.$$

This is a proved rate for the soft-constraint scheme, better than the hard-constraint figure by a factor of two and of the same order: the loss is intrinsic to any argument whose only inputs are the gap and the hopping norm, since their ratio is about 10^{-2} .

2. The analytic-structure route, set up

For the hard constraint, the fluctuation covariance is diagonal in the coarse momentum $\bar{k} \in [-\frac{\pi}{2}, \frac{\pi}{2}]^4$ and couples the sixteen folded fine momenta $k = \bar{k} + \pi n$, $n \in \{0, 1\}^4$. The tube weight has the product form

$$|\hat{w}(k)|^2 = \prod_{\nu \neq \mu} \cos^2 \frac{k_\nu}{2} \cdot \cos^4 \frac{k_\mu}{2}, \quad \hat{k}^2 = \sum_{\nu} 4 \sin^2 \frac{k_\nu}{2},$$

and folding exchanges $\cos^2 \leftrightarrow \sin^2$ in the flipped components. With $s(\bar{k}) = \sum_n |\hat{w}(\bar{k} + \pi n)|^2 / \hat{k}^2(\bar{k} + \pi n)$, the diagonal entry of the fluctuation covariance at the unfolded momentum is

$$C_{\text{fl}}(\bar{k}, \bar{k}) \propto \frac{\sum_{n \neq 0} |\hat{w}(\bar{k} + \pi n)|^2 / \hat{k}^2(\bar{k} + \pi n)}{|\hat{w}(\bar{k})|^2 + \hat{k}^2(\bar{k}) \sum_{n \neq 0} |\hat{w}(\bar{k} + \pi n)|^2 / \hat{k}^2(\bar{k} + \pi n)},$$

in which the massless pole $1/\hat{k}^2(\bar{k})$ has cancelled identically. Three facts fix where singularities can be. (a) For $\bar{k} \rightarrow \bar{k} + i\kappa e$ with $\cosh \kappa < 2$, every flipped denominator satisfies $\text{Re } \hat{k}^2(\bar{k} + \pi n) \geq 2 - 2(\cosh \kappa - 1) > 0$, so only the unfolded term can have a pole, and it has cancelled. (b) $|\hat{w}(\bar{k})|^2 \geq \frac{1}{32}$ throughout the strip, since each factor has modulus at least $\frac{1}{2}$ there. (c) The remaining singularities are the zeros of the denominator $D(\bar{k}) = |\hat{w}(\bar{k})|^2 + \hat{k}^2(\bar{k})R(\bar{k})$, with R the flipped sum, analytic and bounded in the strip. **The decay rate of the hard-constraint fluctuation propagator is the width of the largest strip $|\text{Im } \bar{k}| < \kappa_*$ in which D has no zero**, and κ_* is expected to be of order one. Establishing it is a bounded computation with four real variables and one explicit function, and it is the defined task for an order-one rate; no part of it is done here.

3. The shape of the threshold

Part 2 organizes the remainder $S_W - S_2$, the Haar correction and the non-abelian averaging as a gas of polymers on plaquettes, integrated against the fluctuation Gaussian. Its structure is fixed by three ingredients, whatever the details.

Vertices. The cubic term of $N - \text{Re tr } U_p$ is $\text{tr } (d\theta [\theta, \theta])$ -type; with fluctuations of root-mean-square size $\frac{3}{2}g$ per link (part 1), $d\xi$ is of size up to $6g$ and the cubic vertex per plaquette is of size $\frac{2}{g^2} \cdot 6g \cdot \frac{9}{4}g^2 \cdot c_{\text{col}} \simeq 27c_{\text{col}}g$, so C_3g with C_3 of order 10 to 30; quartic and Haar terms are $O(g^2)$ and subleading.

Covariance sums. Factorizing the Gaussian integral of a product over a set of plaquettes into connected polymers by the Brydges–Kennedy interpolation puts one fluctuation covariance on each edge of a spanning tree, so every polymer of n plaquettes carries a factor $(\|C_{\text{fl}}\|_{1,\kappa}/g^2)^{n-1}$ with

$$\frac{\|C_{\text{fl}}\|_{1,\kappa}}{g^2} = \sup_x \sum_y \frac{|C_{\text{fl}}(x, y)|}{g^2} \leq \frac{9}{2} \left(\coth \frac{\kappa}{2} \right)^4 \simeq \frac{72}{\kappa^4} \quad (\kappa \ll 1),$$

using $|C_{\text{fl}}(x, y)| \leq \frac{9}{2}g^2 e^{-\kappa|x-y|_\infty}$.

Entropy. Plaquette adjacency 20 in four dimensions, so connected sets of n plaquettes through a given one number at most $(20e)^{n-1}$ (Wilson note §2).

The Kotecký–Preiss condition then holds when the per-plaquette activity times e , the adjacency entropy and the covariance sum is below one,

$$e \cdot C_3g \cdot 20e \cdot \frac{72}{\kappa^4} \leq 1 \quad \Longleftrightarrow \quad g \leq g_{\text{pert}} \simeq \frac{\kappa^4}{1.1 \times 10^4 C_3},$$

which is the form quoted in the abstract.

κ (per lattice unit)	origin	g_{pert} ($C_3 = 20$)	g_{pert}^2
7×10^{-4}	proved, hard constraint	10^{-18}	10^{-36}
1.7×10^{-3}	proved, soft constraint	4×10^{-17}	10^{-33}
0.5	expected order-one rate	3×10^{-7}	10^{-13}
1	optimistic	5×10^{-6}	2×10^{-11}

This is a scaling estimate: the constants C_3 , the adjacency, and the tree-graph organization are the standard ones and the form is robust, but no polymer expansion has been carried out here, and a full treatment adds factors of order one to ten in either direction. It does not change the conclusion, because the conclusion is about orders of magnitude.

4. The conclusion for the weak side

The physical onset of the perturbative running is at $g^2 \simeq 1$, $\beta_W \simeq 6$, where asymptotic scaling is observed. The explicit small-field step, done with the sharpest propagator rate one can hope for and the standard organization, covers $g^2 \lesssim 10^{-12}$; with the proven rate, 10^{-35} . Every step between g_{pert}^2 and $g^2 \simeq 1$ is, in the rigorous sense, a step without a small parameter, and their number is irrelevant since none is controlled. The strong side's corresponding deficit, from the proved $\beta_W = 0.0135$ to the physical 5.7, is a factor 400, and its methods are polymer expansions of the same family. The difference is that the strong-side gas has activities $\beta_W/6$ per plaquette and a local structure, while the weak-side gas has activities $C_3 g$ dressed by covariance sums that scale like κ^{-4} and vertex constants of order ten.

So H1a, the sharpness sub-target, is closed: sharpening constants within the present method cannot bring g_{pert}^2 within twelve orders of magnitude of the physical onset, and the meeting of the two reaches through constant-chasing is not a viable route. H1 is a methods problem, which is the content of the constructive programme extended through the confinement scale: control of the renormalization steps at couplings where the fluctuation is not small compared with the nonlinearity, with no expansion in any parameter. That is the open problem, stated as precisely as this programme can state it.

5. Consequence for STATE

The decay rate is proved at 10^{-3} per lattice unit by two routes and its order-one value is reduced to a zero-free strip for the explicit function D of Section 2. The threshold of the remainder gas scales as κ^4 and lies at $g^2 \sim 10^{-12}$ even for an ideal rate, twelve orders below the physical onset of the running. The weak-side sharpness sub-target is closed as not viable; the open problem is a methods problem at order-one coupling, and the map of this programme ends there.